

CLOUD COMPUTING COURSE

Chapter1: Introduction to Cloud Computing

ABSTRACT
Chapter1 introduces cloud-computing definition, characteristic, Models and history.

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Introduction to Cloud Computing

-Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

This means:

That instead of all the computer hardware and software you're using sitting on your desktop, or somewhere inside your company's network, it's provided for you as a service by another company and accessed over the Internet, usually in a completely seamless way. Exactly where the hardware and software is located and how it all works doesn't matter to the user—it's just somewhere up in the "cloud" that the Internet represents.

Example: Yahoo, Gmail, Hotmail: instead of running an e-mail program on your computer, you log in to a web e-mail account remotely. The software and storage for your account doesn't exist on your computer, it is on the services computer cloud.

-Allows us to use IT services as conveniently as using utilities like water and electricity.

-Ready to use IT services over the internet.

Cloud Computing Examples

Examples: AWS, Azure, Google Cloud and HUWAEI CLOUD user portal-Products.

Example2: Cloud Computing Applications such as:

1. Backup and Restore application
2. Google Translate
3. iReader
4. Gmail
5. SoundCloud: You can build up a personal collection of tracks you like and access them from any device, anytime, anywhere.

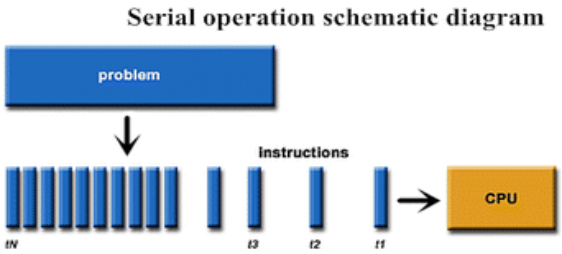
Computing History

1. Serial Computing

A processing in which one task is completed at a time and all the tasks are run by the processor in a sequence. This type of processing is also known as **sequential processing**.

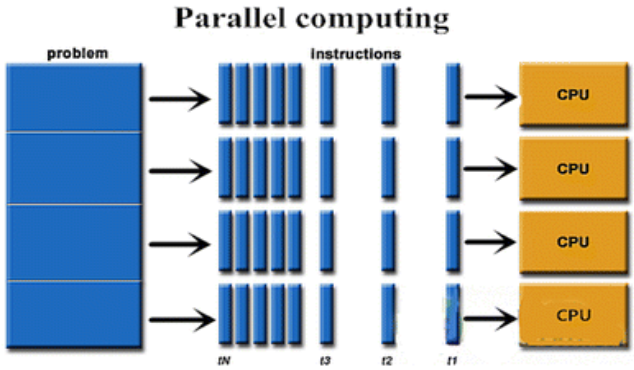
Example: Any operating system running on the single processor is an example of the serial operating system.

Pentium 3 and Pentium 4 are serial processing computers. Windows 95 and Windows 98 are examples of operating systems, which do the serial processing.



2. Parallel Computing

A type of processing in which multiple tasks are completed at a time by different processors.it consists of multiple processors that communicate with each other using a shared memory.

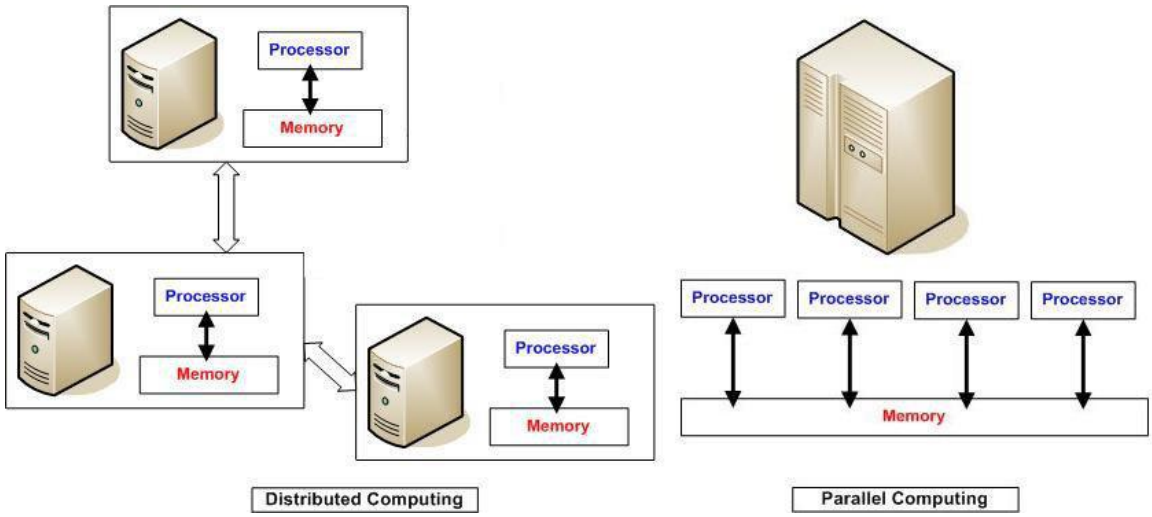


Example: An operating system running on the multicore processor is an example of the parallel operating system. Windows 7, 8, 10 are examples of operating systems which do parallel processing. In today life all latest operating systems support parallel processing.

3. Distributed Computing

It is a type of computing in which a group of computers are linked together so that they can act as a single entity, so all computers work together to achieve a common goal. In distributed systems, there is no shared memory and computers communicate with each other through message passing.

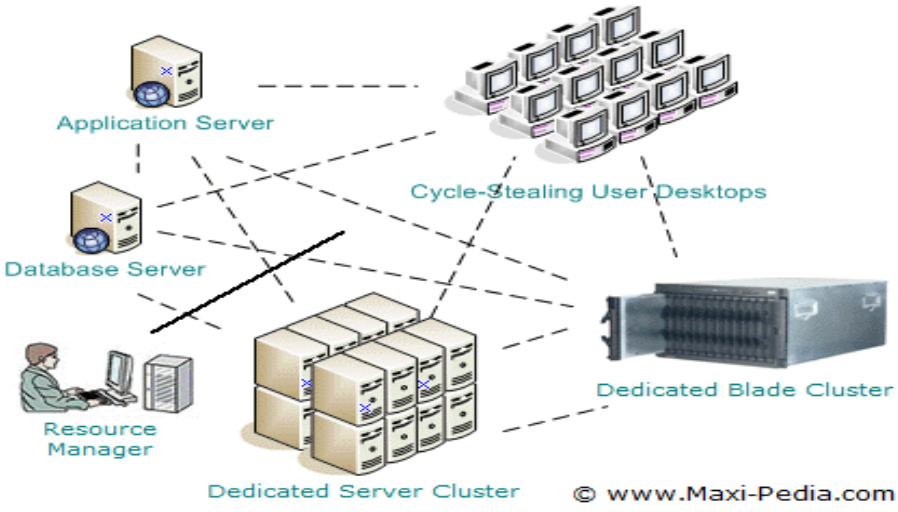
A computer in the distributed system is a node while a collection of nodes is a cluster.



4. Grid Computing

It is a collection of **computer resources** from multiple locations to reach a common goal. grid works on various tasks within a network, It is designed to solve problems that are too big for a supercomputer. It is often used in structural analysis, Web services such as ATM banking, back-office infrastructures, and scientific or marketing research.

The key distinction between distributed computing and grid computing is mainly the way resources are managed: Distributed computing uses a centralized resource manager and all nodes cooperatively work together as a single unified resource or a system while **Grid computing** utilizes a structure where each node has its own resource manager and the system does not act as a single unit.



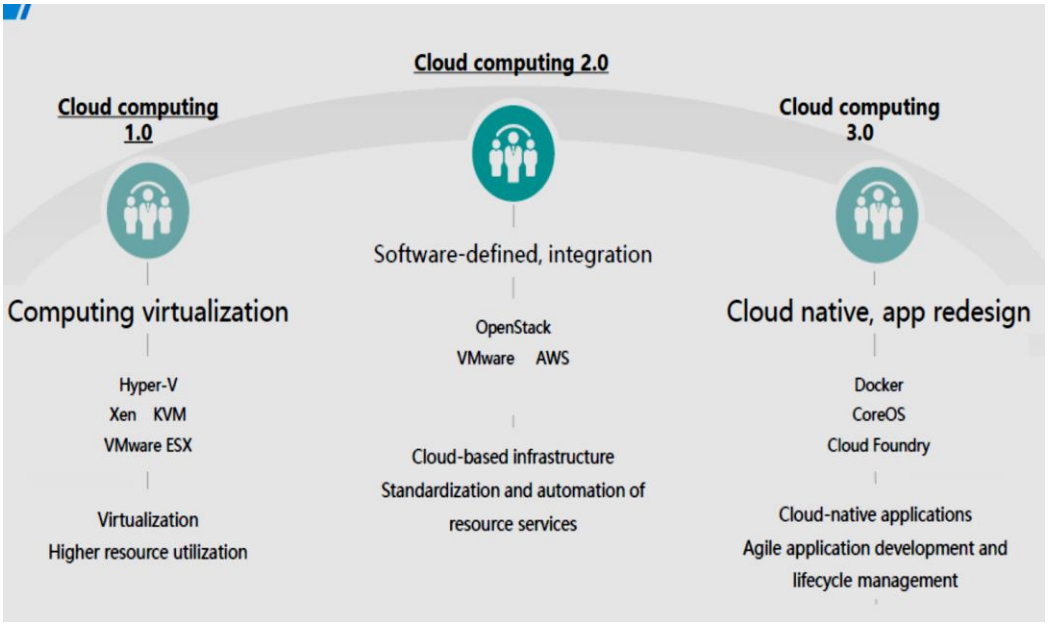
5. Cloud Computing

It is a type of computing that relies on sharing computer resources rather than having a local servers or personal devices to handle applications. **Parallel computing, grid computing, and Distributed computing all belongs to cloud computing.**

Internet History

ARPANET was born→ **TCP/IP** protocol was established→**DNS** introduced → **WWW** open to public.

Cloud Computing Evolution



Cloud Computing Characteristic

1. On-Demand Self-Service.

After determining the required cloud computing services, users can apply for cloud computing resources by themselves so:

1. The engineer purchased a cloud server from Huawei's public cloud according to his own needs
2. The engineer upgrades the cloud server CPU he purchased from 2 to 4.
3. The engineer changes the operating system of the cloud server he purchased from Linux to Windows

Perquisites of on-demand self-service:

1. Consumers understanding of their own needs and what product accommodates such needs.
2. Resource Pooling
3. Network connectivity

4. Broad Network Access.

- **Through the network, users can use self-purchased cloud computing resources through different terminals, at any time and in different locations.**
- **Through the network, users can view the status of self-purchased cloud computing resources at any time**

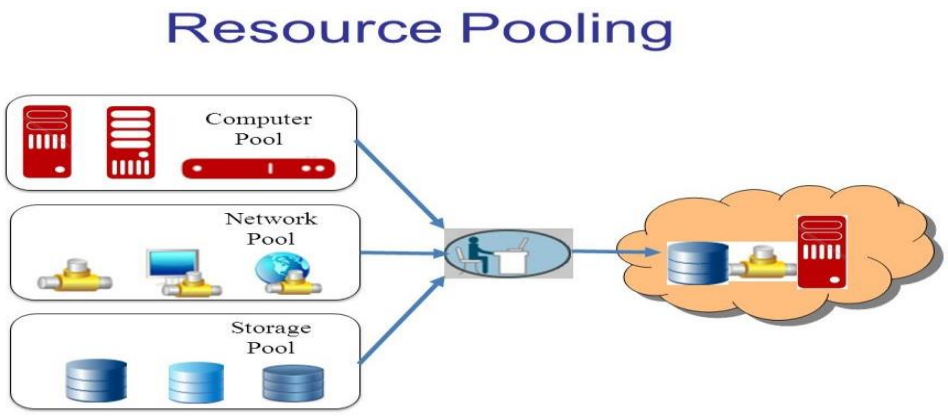
5. Resource Pooling.

Resource pooling means that multiple customers are serviced from the same physical resources (**compute, storage and network resources**). Providers' resource pool should be very **large and flexible** enough to service multiple client requirements

The importance of resource pooling:

1. Allow you to have the exact amount of resources you need.
2. Shield the differences in the under lying resources.
3. Provide maximum utilization of the resources.

Examples of resources include storage, processing, memory, and network bandwidth.



2. Rapid Elasticity.

Elasticity is a landmark of cloud computing and it implies that customers can rapidly **provision and de-provision** any of the cloud computing resources. Rapid provisioning and de-provisioning might apply to storage or virtual machines or customer applications.

Scaling can be:

3. **Horizontal Scale**: this done by increasing number of servers.
4. **Vertical Scale:** increasing number of resources within the same server.

5. Measured service.

This means that cloud resource usage (whether virtual server instances that are running, storage in the cloud, bandwidth, and active user accounts) is monitored, measured and reported by the cloud service provider, providing transparency for both the provider and consumer of the utilized service.

Measuring is based on different criteria's:

- Duration of usage.

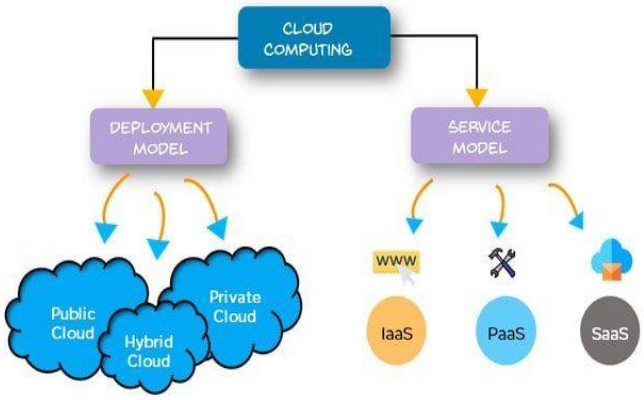
- Resource quota

- Volume of data transmitted.

Cloud Computing Models

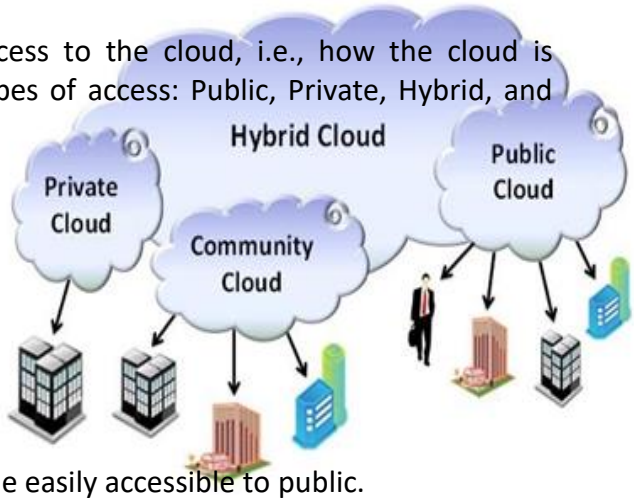
There are certain services and models working behind the scene making the cloud computing feasible and accessible to end users. Following are the working models for cloud computing:

- Deployment Models
- Service Models



1. Cloud Computing Deployment Models

Deployment models define the type of access to the cloud, i.e., how the cloud is located? Cloud can have any of the four types of access: Public, Private, Hybrid, and Community.



1. Public Cloud

Public Cloud allows systems and services to be easily accessible to public.

The public cloud is generally constructed by carriers or cloud service providers. It sells or leases services to the public.

The IT giants such as **Google**, **Amazon** and **Microsoft** offer cloud services via Internet.

2. Private Cloud

Private Cloud allows systems and services to be accessible within an organization. The Private Cloud is operated only within a single organization. However, it managed internally by the organization itself or by third-party. ***The private cloud is deployed on the premises of enterprises and used by internal employees.***

Private cloud providers such as **Microsoft**, **VMware**

3. Hybrid Cloud

The hybrid cloud is a mixture of public and private cloud, in which the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.

4. Community Cloud

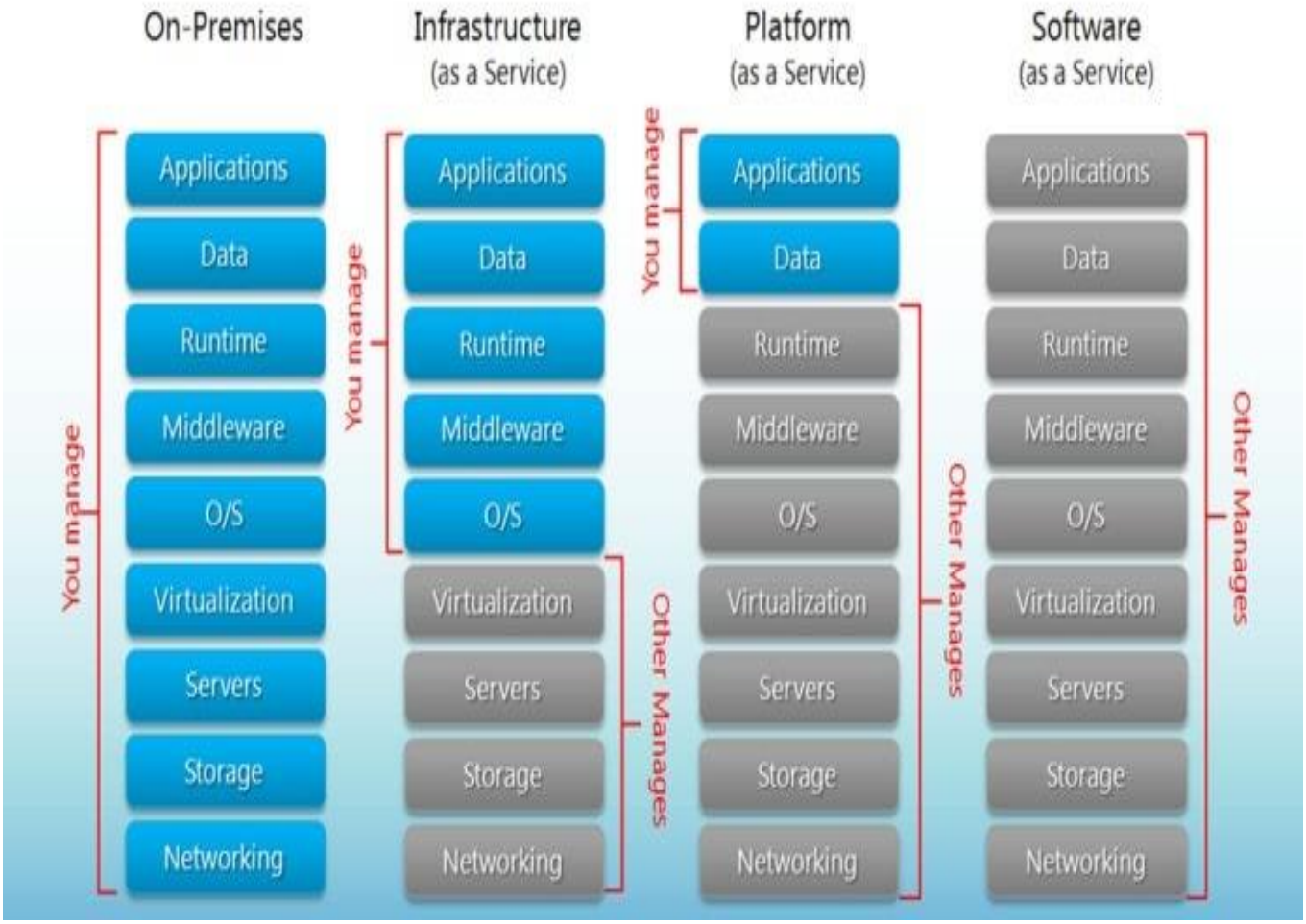
It allows system and services to be accessible by group of organizations. It shares the infrastructure between several organizations from a specific community. It may be managed internally by organizations or by the third-party.

The industry cloud generally has attributes of a certain industry, which is a "large private cloud" or "small public cloud".

2. Cloud Computing Service Models

Cloud computing is based on service models. These are categorized into three basic service models, which are:

- **Infrastructure-as–a-Service (IaaS)**
- **Platform-as-a-Service (PaaS)**
- **Software-as-a-Service (SaaS)**



1. **Infrastructure as a service (IaaS):** is an instant computing infrastructure, provisioned and managed over the internet. IaaS quickly scales up and down with demand, letting you pay only for what you use. It helps you avoid the expense and complexity of buying and managing your own physical servers and other datacenter infrastructure.

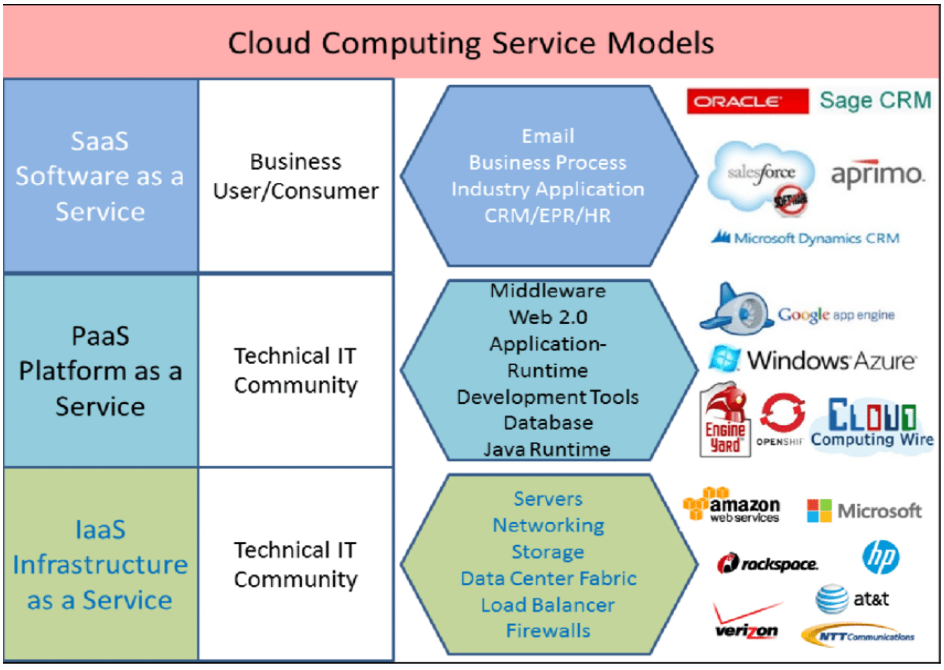
The Management of IaaS: A cloud computing service provider manages the infrastructure, while you purchase, install, configure, and manage your own software, operating systems, middleware, and applications.

2. **Platform as a service (PaaS)** is computing model where cloud service providers deliver a set of development kits and programming environments as well as prebuilt application components. These prebuilt components often include back-end frameworks, libraries, database management tools, and other components. Like IaaS, PaaS includes infrastructure (servers, storage, and networking) but also middleware, development tools, database management systems, and more.

   The Management of PaaS: You manage the applications and services you develop, and the cloud service provider typically manages everything else.

3. **Software as a service (SaaS):** allows users to connect to and use cloud-based apps over the Internet. All of the underlying infrastructure, middleware, app software, and app data are located in the service provider's data center. Common examples are email, calendaring, and office tools (such as Microsoft Office 365).

   The Management of SaaS: The service provider manages the hardware and software.



Cloud Computing Architecture

Cloud computing comprises of two components:

1. Front End: it consist client part of cloud computing system (PC, Laptop or Mobile). It comprise of **interfaces and applications** that are required to access the cloud-computing platform.

2. Back End: refers to the cloud itself, it comprises of the resources that are required for cloud computing services. It consists of **virtual machines, servers, data storage, security mechanism etc.** It is under provider's control.

